Chapter 1

Conventions and notation

This chapter fixes the symbols, the typographical rules and the index conventions that the rest of the compendium and the paper share. Everything here is a decision, not a result. The decisions were taken once, in the file `paper/notation.tex`, and every chapter of this compendium is written against that file. A reader who wants to check the paper line by line should read this chapter first, because several of the paper's choices differ from the habits of the diffusion literature, and the differences are deliberate.

1.1 How to read this compendium

Every chapter is self-contained at the level of a chapter: it defines the objects it uses, derives every formula it states (the scalar case in full, then the multivariate statement), and works one small example by hand. Four typographical devices recur.

- A **term** in bold at its first use is a definition; the term is also entered in the index at the end of the document.
- A paragraph opening with *Where it is used* closes each chapter and names the paper sections, the experiment identifiers (E01 to E21) and the code paths in which the chapter's objects appear.
- A paragraph opening with *Why it matters here* explains, inside a derivation, why a particular hypothesis or constant is the one the paper needs.
- A path set in typewriter, such as `src/markov_score/theory.py`, is a file of the repository. Every such path was checked to exist when the chapter was written.

Theorems, propositions, lemmas and corollaries are numbered within chapters, as are definitions, examples, assumptions and remarks. "The paper" always means the ICML manuscript *What Survives Diffusion Noise?*; its sections, theorems and figures are referred to in words ("the paper, Section 3, Theorem 2"), never by cross-reference, because the two documents are compiled separately. Experiments are referred to by their registry identifier (E01, ..., E21), which names an entry of `experiments/registry/`. No measured number from the experiments is typed in this compendium: those numbers live in generated macros of the paper, and the compendium says "see the paper, Figure 2" instead. Analytic constants such as $\frac{1}{6}$, $\frac{1}{24}$ or e^{-4t} are typed freely.

1.2 The symbol table

Table 1.1 lists every macro of `paper/notation.tex` and every macro added in the compendium's preamble, with its rendering, its meaning and the chapter in which the object is defined from scratch. The macro names are given so that a reader of the sources can search for them.

Table 1.1: Every macro of the paper's notation file and of the compendium preamble.

macro	renders as	meaning	defined in
Diffusion			
`\Pz`	P_0	the clean law of a sequence, a density on $\mathbb{R}^L$	Markov chains
`\Pt`	P_t	the noised law at diffusion time t, the law of $\mathbf{x}_t$	The OU channel and Tweedie
`\Dt`	Δ_t	$1 - e^{-2t}$, the conditional variance of one noised coordinate	The OU channel and Tweedie
`\score`	S	the diffusion score $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$	The OU channel and Tweedie
`\gwn`	η	standardised Gaussian white noise, $\langle \eta_i(t)\eta_j(t')\rangle = \delta_{ij}\delta(t - t')$	The OU channel and Tweedie
`\tmin`, `\tmax`	$t_{\min}$, $t_{\max}$	the ends of the noise schedule	The protocol
`\pmean{·}`	$\langle \mathbf{a} \rangle_{x,t}$	posterior mean $\mathbb{E}[\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}]$, in bracket notation	The OU channel and Tweedie
`\pmeansite{i}`	$\langle a_i \rangle_{x,t}$	posterior mean of one site	The OU channel and Tweedie
Vectors and matrices			
`\vect{v}`, `\mat{A}`	$\mathbf{v}$, $\mathbf{A}$	bold lowercase for vectors, bold uppercase for matrices and tensors	this chapter
`\va`	$\mathbf{a}$	the clean chain, a point of $\mathbb{R}^L$; its sites are the scalars a_i	Markov chains
`\vx`	$\mathbf{x}$	the noised chain, $\mathbf{x} = \mathbf{x}_t \in \mathbb{R}^L$	The OU channel and Tweedie
`\vz`	$\mathbf{z}$	a standard normal draw in $\mathbb{R}^L$	Gaussian integrals and Hermite
`\Sig`	$\mathbf{\Sigma}$	a covariance matrix; $\mathbf{\Sigma}_0$ is the clean chain's, $\mathbf{\Sigma}_t = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}$ the noised one's	Markov chains
`\Id`	$\mathbf{I}$	the identity matrix	this chapter
Chain and graphical model			
`\nsites`	L	chain length, equal to the ambient dimension	Markov chains
`\nseq`	M	number of observed sequences	Markov chains
`\ngrid`	N_{g}	number of quadrature grid points	Grid quadrature
`\gridpt{k}`	u_k	the k-th grid point	Grid quadrature
`\gridh`	h	the grid spacing	Grid quadrature
`\halfwidth`	A	the grid half-width; the domain is $[-A, A]$	Grid quadrature
`\kernel`	K	the Markov transition kernel $K(a' \mid a)$	Markov chains
`\kernelth`	K_θ	a transition kernel with parameters θ	Maximum likelihood and EM
`\like`	ℓ_t	the unary likelihood factor $\ell_t(x_i \mid a_i)$	Factor graphs and BP
`\lmsg`, `\rmsg`	$\overrightarrow{m}$, $\overleftarrow{m}$	forward and backward messages	Factor graphs and BP
`\innov`	φ	the innovation density	Markov chains
`\corr`	ρ	the autoregression coefficient	Markov chains
`\ivar`	q	the innovation variance, $q = 1 - \rho^2$ in the paper	Markov chains
`\ncomp`	C	number of mixture components of the fitted innovation	Maximum likelihood and EM
Estimation			
`\params`	θ	a parameter vector	Fisher information
`\Qfun`	$\mathcal{Q}$	the EM auxiliary function	Maximum likelihood and EM
`\Xis`	$\mathbf{\Xi}$	the expected transition statistic, a matrix on the grid	Maximum likelihood and EM

macro	renders as	meaning	defined in
`\Lik`	$\mathcal{L}$	the marginal log-likelihood	Maximum likelihood and EM
`\mmse`, `\lmmse`	$\widehat{\mathbf{a}}_{\mathrm{MMSE}}$, $\widehat{\mathbf{a}}_{\mathrm{LMMSE}}$	the posterior mean and the linear MMSE estimator	Gaussian closure and LMMSE
Operators			
`\E`, `\Prob`	$\mathbb{E}$, $\mathbb{P}$	expectation, probability	Moments and cumulants
`\Reals`	$\mathbb{R}$	the real line	this chapter
`\Cov`, `\Var`	Cov, Var	covariance, variance	Moments and cumulants
`\KL`	D_{KL}	Kullback–Leibler divergence	Maximum likelihood and EM
`\normal{m}{v}`	$\mathcal{N}(m, v)$	the Gaussian law with mean m and variance v	Gaussian integrals and Hermite
`\Pgauss`	Π_{G}	Gaussian moment projection: the Gaussian with the argument's mean and variance	Gaussian closure and LMMSE
`\dd`	d	the differential in integrals, $\mathrm{d}x$	this chapter
Pending-data guard and repository pointers			
`\needsdata`	red bracket	with an argument naming the slot: marks a place in the paper still waiting for a frozen number; the build fails if any survives	this chapter
`\codepath`, `\codefile`	typewriter	a repository path	this chapter
`\coderef{p}{f}`	path → name	a path with a function or target inside it	this chapter
Macros the paper defines in `paper/main.tex`			
`\Frob{T}`	$\|\mathbf{T}\|_F$	Frobenius norm, the root of the sum of squares over ordered indices	Moments and cumulants
`\Hen{k}`	He_k	the k-th probabilists' Hermite polynomial	Gaussian integrals and Hermite
`\kfour`	$\boldsymbol{\kappa}_4$	the fourth-cumulant tensor of the clean law	Moments and cumulants
`\Sgauss`	S_{G}	the score of the covariance-matched Gaussian twin, $-\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$	Gaussian closure and LMMSE
`\tstar`	t_*	the crossover time $\frac{1}{2}\log\lambda_{\max}(\boldsymbol{\Sigma}_0)$	Markov chains; Attenuation theorems
`\lmax`	$\lambda_{\max}$	the largest eigenvalue of a symmetric matrix	Markov chains
`\Pgaussl`	P_{G}	the Gaussian law with the clean chain's covariance (the twin)	Gaussian closure and LMMSE
Compendium-only macros			
`\tr`, `\diag`	tr, diag	trace; diagonal matrix built from a vector	this chapter
`\argmax`, `\argmin`	arg max, arg min	maximiser and minimiser	this chapter
`\term{·}`	bold	a defined term, also indexed	this chapter
`\where{·}`, `\why{·}`	italic lead-in	the two closing paragraphs described above	this chapter
`\code{·}`	typewriter	a repository path or identifier	this chapter

Three macros are typeset slightly differently in the paper and in this compendium, and the reader should not be surprised by the difference. The paper writes the fourth-cumulant tensor as κ_4 in light face, the compendium as $\boldsymbol{\kappa}_4$ in bold, because the compendium's rule is that tensors are bold. The paper writes the twin's score with a superscript, S^{G}, the compendium with a subscript, S_{G}. The paper writes the crossover as t_* and the compendium as t_*. The objects are

the same.

1.3 Indices: sites, sequences, grid points, and why n and d are not used

Sites are the coordinates of a sequence. A clean sequence is $\mathbf{a} = (a_1, \dots, a_L)$, and i, j always index sites, $i \in \{1, \dots, L\}$. The **chain length** L is therefore also the **ambient dimension** of the space in which the diffusion runs: the noised sequence $\mathbf{x}_t$ lives in $\mathbb{R}^L$, the score is a map $\mathbb{R}^L \times [0, \infty) \to \mathbb{R}^L$, and every covariance matrix is $L \times L$. The paper writes this identification once, as $n = L$, and then never uses n again.

Sequences are the independent draws. The data are M sequences $\mathbf{a}^1, \dots, \mathbf{a}^M$ (or their noised versions), and μ indexes sequences, $\mu \in \{1, \dots, M\}$. Fisher information is stated *per sequence*, and the Cramér–Rao and local asymptotic minimax statements are in the limit $M \to \infty$ at fixed L and fixed t.

Grid points are the abscissae on which messages are stored. There are N_g of them, $u_1, \dots, u_{N_\mathrm{g}}$, uniformly spaced by h on $[-A, A]$, and k, l index them. A matrix on the grid, such as the discretised kernel $K(u_k \mid u_l)$ or the expected transition statistic $\mathbf{\Xi}_{kl}$, is $N_\mathrm{g} \times N_\mathrm{g}$ and is indexed *child first, parent second*: k is the site a_i, l is the site a_{i-1}. This orientation is the one thing a self-consistent recursion cannot catch if it is wrong, which is why `tests/test_fisher_bruteforce.py` checks it against an enumeration that shares nothing with the recursion.

Multi-indices $\nu = (\nu_1, \dots, \nu_L) \in \mathbb{N}^L$ label Hermite products; the **degree** is $|\nu| = \sum_i \nu_i$ and $\nu! = \prod_i \nu_i!$.

Why n and d are avoided. Much of the diffusion literature writes n for the number of samples and d for the dimension of the data, and much of the statistics literature writes n for the sample size as well. In this laboratory the dimension *is* the chain length, and the reader should never have to remember that a statement "in dimension d" is a statement about a chain of d sites; so the dimension is called L and the number of sequences M. The letter n is kept for the degree of a scalar Hermite polynomial He_n and for nothing else. Two further collisions were removed at the same time: the transition kernel is K and only K, and the number of grid points is N_g, whereas an earlier draft used K for the grid size and W for the kernel.

Levels. The noise schedule is a finite set of diffusion times, twelve of them log-spaced on $[t_\mathrm{min}, t_\mathrm{max}]$ in the frozen protocol; "level" means one such t. A statistic "pooled over the schedule" sums over levels before taking a root or a ratio; see the chapter on the statistics of the experiments.

1.4 Vectors, matrices, tensors and scalars

Vectors are bold lowercase ($\mathbf{a}$, $\mathbf{x}$, $\mathbf{z}$, $\mathbf{s}$), matrices and tensors bold uppercase ($\mathbf{\Sigma}$, $\mathbf{I}$, $\mathbf{\Xi}$, $\boldsymbol{\kappa}_4$). Anything indexed down to a scalar loses its bold: a_i is the i-th site of $\mathbf{a}$, Σ_{ij} an entry of $\mathbf{\Sigma}$, $\kappa_4[ijkl]$ an entry of $\boldsymbol{\kappa}_4$. The Euclidean norm of a vector is $|\mathbf{a}|$ or $\|\mathbf{a}\|$; the Frobenius norm of a matrix or of a symmetric tensor is $\|\mathbf{T}\|_F$, with

$$\|\mathbf{T}\|_F^2 = \sum_{i_1, \dots, i_r} T_{i_1 \cdots i_r}^2, \tag{1.1}$$

the sum running over *ordered* index tuples. For a symmetric matrix this counts each off-diagonal entry twice, and for a symmetric 4-tensor it counts an entry with distinct indices 24 times. The paper's constants $\frac{1}{2}\|\partial_\theta \mathbf{\Sigma}_0\|_F^2$ and $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$ are stated with this convention, and `src/markov_score/theory.py` sums the full (L, L, L, L) array; the chapter on moments and cumulants explains how the ordered-index sum turns into a sum over multisets with weights $4!/\nu!$, which is where the paper's $\frac{1}{24}$ comes from.

$\mathbf{I}$ is the identity, tr the trace, $\mathrm{diag}(\mathbf{v})$ the diagonal matrix with $\mathbf{v}$ on the diagonal, $\mathbf{A} \succ 0$ positive definiteness, $\lambda_{\max}(\mathbf{A})$ the largest eigenvalue. $\mathcal{N}(m, v)$ is the Gaussian law with mean m and variance (or covariance matrix) v. The symbol γ is reserved for the standard Gaussian density on $\mathbb{R}^L$, and γ_{Δ_t} for $\mathcal{N}(0, \Delta_t \mathbf{I})$.

1.5 Hermite polynomials: probabilists' versus physicists'

Two families of polynomials carry the name Hermite, and they differ by a rescaling that changes every constant. The **probabilists' Hermite polynomials** are

$$\mathrm{He}_n(x) = (-1)^n e^{x^2/2} \frac{\mathrm{d}^n}{\mathrm{d}x^n} e^{-x^2/2}, \tag{1.2}$$

so that $\mathrm{He}_0 = 1$, $\mathrm{He}_1 = x$, $\mathrm{He}_2 = x^2 - 1$, $\mathrm{He}_3 = x^3 - 3x$, $\mathrm{He}_4 = x^4 - 6x^2 + 3$; they are monic and orthogonal under the standard Gaussian, with $\mathbb{E}_\gamma[\mathrm{He}_n \mathrm{He}_m] = n!\, \delta_{nm}$. The **physicists' Hermite polynomials** are $H_n(x) = (-1)^n e^{x^2} \frac{\mathrm{d}^n}{\mathrm{d}x^n} e^{-x^2}$, with leading coefficient 2^n and orthogonal under e^{-x^2} with norm $2^n n! \sqrt{\pi}$. The two are related by

$$H_n(x) = 2^{n/2} \mathrm{He}_n(\sqrt{2}\, x), \qquad \mathrm{He}_n(x) = 2^{-n/2} H_n(x/\sqrt{2}). \tag{1.3}$$

The paper and this compendium use the probabilists' family throughout, because the noise is standardised and the orthogonality weight is then exactly the law of the noise. One notational hazard must be named: the paper writes $H_\nu(\mathbf{x}) = \prod_i \mathrm{He}_{\nu_i}(x_i)$ for the *multivariate product* of probabilists' polynomials. That H_ν is not a physicists' polynomial, despite the capital letter; it is the product of probabilists' ones, and $\mathbb{E}_\gamma[H_\nu H_\mu] = \nu!\, \delta_{\nu\mu}$. The chapter on Gaussian integrals and Hermite polynomials derives all of this.

1.6 Standardised white noise and the explicit $\sqrt{2}$

The forward process is written, in the paper and in the code (`src/markov_score/noising.py`), as the Ornstein–Uhlenbeck equation

$$\mathrm{d}\mathbf{x}_t = -\mathbf{x}_t \, \mathrm{d}t + \sqrt{2} \, \mathrm{d}\mathbf{W}_t, \qquad \mathbf{x}_0 = \mathbf{a} \sim P_0, \tag{1.4}$$

with $\mathbf{W}$ a standard Brownian motion, equivalently with a **standardised white noise** η satisfying $\langle \eta_i(t) \eta_j(t') \rangle = \delta_{ij} \delta(t - t')$ and $\mathrm{d}\mathbf{W}_t = \eta(t)\, \mathrm{d}t$. The factor $\sqrt{2}$ is carried explicitly for one reason: with it, the stationary variance of each coordinate is exactly one. Solving Equation (1.4) by variation of constants,

$$\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{2} \int_0^t e^{-(t-s)} \, \mathrm{d}\mathbf{W}_s, \qquad \mathrm{Var}\Big(\sqrt{2} \int_0^t e^{-(t-s)} \, \mathrm{d}W_s\Big) = 2 \int_0^t e^{-2(t-s)} \, \mathrm{d}s = 1 - e^{-2t}, \tag{1.5}$$

so that, conditionally on $\mathbf{a}$,

$$\mathbf{x}_t \mid \mathbf{a} \;\sim\; \mathcal{N}\Big(e^{-t}\mathbf{a}, \Delta_t \mathbf{I}\Big), \qquad \Delta_t = 1 - e^{-2t}, \qquad \text{that is} \qquad \mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\, \mathbf{z}, \;\; \mathbf{z} \sim \mathcal{N}(0, \mathbf{I})\,. \tag{1.6}$$

Since $e^{-2t} + \Delta_t = 1$, a coordinate of unit variance keeps unit variance at every t, and $P_t \to \mathcal{N}(0, \mathbf{I})$. Two other conventions appear in the literature and are *not* used here. The physicists' convention absorbs the $\sqrt{2}$ into the noise by demanding $\langle \eta(t) \eta(t') \rangle = 2\delta(t - t')$; the formulas are the same but the noise is no longer standardised. The "variance-preserving" convention of the score-based literature writes $\mathrm{d}\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}\, \mathrm{d}t + \sqrt{\beta(t)}\, \mathrm{d}\mathbf{W}$ with a schedule $\beta(t)$; Equation (1.4) is the case $\beta \equiv 2$, and the noise level is then parameterised by t directly. The decay factor is written e^{-t} in every displayed equation of the paper and is never abbreviated to α_t; inside proofs the paper allows the local abbreviation $\alpha = e^{-t}$, always introduced in the sentence that uses it, and this compendium does the same. The **signal-to-noise ratio** of the channel is e^{-2t}/Δ_t, which is $e^{-2t}(1 + O(e^{-2t}))$ at large t; this is why the paper's expansions are in powers of e^{-2t}.

1.7 What "the score" means in the paper

The word **score** is used in two different senses in statistics and in diffusion modelling, and the paper is careful to use it in one of them. The **diffusion score** is the gradient of the log-density of the *noised data law* with respect to the *data*,

$$S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x}), \tag{1.7}$$

a vector field on $\mathbb{R}^L$ for each t. The **statistical score** is the gradient of a log-likelihood with respect to a *parameter*, $\partial_\theta \log p_\theta(\mathbf{x})$, a function of θ for fixed data. The chapter on Fisher information and the Cramér–Rao bound defines both and keeps them apart; the paper writes "the score" for the diffusion score and "the likelihood score" or "per-sequence scores obtained by Fisher's identity" for the statistical one.

Within the diffusion sense, four objects have to be distinguished, and the paper's sentence "throughout, 'the score' means this tree-exact inference, evaluated by validated quadrature" names one of them. They are:

(L1) exact functional sum-product on the posterior factor graph, which for this model is a chain and therefore a tree, so that the forward and backward messages are exact and Tweedie's identity $S(\mathbf{x}, t) = -(\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t})/\Delta_t$ turns the exact posterior mean into the exact score;

(L2) Gaussian-projected sum-product, in which every message is replaced by the Gaussian with its first two moments; the paper's Proposition 1 shows this returns the linear MMSE estimator of the covariance twin;

(L3) the numerical implementation of (L1) on a grid of N_{g} points on $[-A, A]$ with trapezoidal weights, at the frozen setting $N_{\mathrm{g}} = 401$, $A = 8$, validated by refinement and by an independent brute-force enumeration on short chains;

(L4) the LMMSE estimator of the true covariance, $e^{-t}\boldsymbol{\Sigma}_0 \boldsymbol{\Sigma}_t^{-1}\mathbf{x}$, and its score $S_{\mathrm{G}} = -\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$, which is what (L2) computes.

"The score" in the paper is (L1) computed as (L3). It is the reference against which every estimator, whether the EM-fitted kernel or a trained network, is scored; the estimand of the experiments is the relative error to this reference, pooled over the schedule. It is never the output of a learned model; those are "estimates", "fits" or "arms". The grid setting is not inherited from the literature but measured: the chapter on grid quadrature explains what truncation and quadrature errors are and how the paper controls them, and why a heavy-tailed innovation tests the domain A rather than the spacing h.

1.8 Other fixed choices

- The clean chain has $a_1 \sim \mathcal{N}(0, 1)$, $|\rho| < 1$, innovation variance $q = 1 - \rho^2$, so that $\mathrm{Var}(a_i) = 1$ and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ for every innovation law. The frozen configuration is $\rho = 0.85$, $L = 32$, Laplace innovations; every departure from it is stated where it occurs.
- Innovation families are always *variance-matched*: each is parameterised by its variance q, and the remaining parameter, if any, is a shape parameter (the Student-t degrees of freedom ν, the mixture separation κ, the generalised-Gaussian exponent β).
- Cumulants are written κ_m; the excess kurtosis of a scalar law is κ_4/κ_2^2; the fourth-cumulant tensor of a vector is $\boldsymbol{\kappa}_4$.
- log is the natural logarithm; a "log-slope" is a derivative or a regression coefficient of $\log y$ against t, so that $y \propto e^{-8t}$ has log-slope -8.
- British spelling and the serial "and" are used; no number is typed from a measurement.

Where it is used. Every chapter. The symbol file is `paper/notation.tex`, read verbatim by `paper/main.tex` and by `reports/compendium/compendium.tex`; the channel convention is implemented in `src/markov_score/noising.py` (the functions `alpha_delta` and `ou_noise_sample`); the Hermite convention in `src/markov_score/theory.py` (the function `he`); the frozen configuration in `experiments/frozen_config.py` and the comparison protocol in `experiments/protocol.py`. The paper's Section 2 fixes the chain and the channel, its Section 3 the Hermite notation, and its Appendix B the grid.

References. Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon and B. Poole (2021), "Score-based generative modeling through stochastic differential equations", ICLR. M. Mézard and A. Montanari (2009), *Information, Physics, and Computation*, Oxford University Press. G. Szegő (1975), *Orthogonal Polynomials*, 4th edition, American Mathematical Society (Chapter V for the two Hermite conventions). C. W. Gardiner (2009), *Stochastic Methods*, 4th edition, Springer (Chapter 4 for the Ornstein–Uhlenbeck process and the white-noise conventions). B. Efron (2011), "Tweedie's formula and selection bias", Journal of the American Statistical Association 106, 1602–1614.